🔌 EduTrack — API Documentation

Academic Management Platform

VTU Internship 2026 — Team 15

Overview

All APIs return JSON responses. Every response includes a `success` boolean field.

Base URL (Local): `http://localhost:8080`

Base URL (Production): `https://edutrack-backend-o1ym.onrender.com`

1. Authentication APIs

1.1 Admin Login

METHOD	POST
ENDPOINT	`/api/auth/login/admin`
DESCRIPTION	Authenticate admin user

```
{
  "email": "admin@edutrack.com",
  "password": "Admin@123"
}
```

1.2 Student Login

METHOD	POST
ENDPOINT	`/api/auth/login/student`
DESCRIPTION	Authenticate student user

```
{
  "email": "student@college.edu",
```

```
  "password": "Test@123"
}
```

1.3 Teacher Login

METHOD	**POST**
ENDPOINT	`/api/auth/login/teacher`
DESCRIPTION	Authenticate teacher user

```
{
  "email": "teacher@college.edu",
  "password": "Test@123"
}
```

1.4 Student Registration

METHOD	**POST**
ENDPOINT	`/api/auth/register/student`
DESCRIPTION	Self-registration for students (blocked if registrationOpen=false)

```
{
  "name": "Arjun Kumar",
  "email": "arjun@college.edu",
  "password": "Test@123",
  "department": "Computer Science and Engineering",
  "year": "1st Year",
  "semester": "Sem 1",
  "phone": "9876543210",
  "usn": "1VT25CS001"
}
```

1.5 Admin Add Student

METHOD	**POST**
ENDPOINT	`/api/auth/admin/add-student`
DESCRIPTION	Admin adds student (bypasses registration gate)

Request Body: Same as student registration

1.6 Admin Add Teacher

METHOD	POST
ENDPOINT	`/api/auth/admin/add-teacher`
DESCRIPTION	Admin adds teacher

```
{
  "name": "Prof. Sharma",
  "email": "sharma@college.edu",
  "password": "Test@123",
  "department": "Computer Science and Engineering",
  "phone": "9876543210",
  "employeeId": "EMP001"
}
```

1.7 Change Admin Password

METHOD	POST
ENDPOINT	`/api/auth/change-password/admin`
DESCRIPTION	Change admin password (requires current password)

```
{
  "id": 1,
  "currentPassword": "Admin@123",
  "newPassword": "NewPass@123"
}
```

1.8 Change Student Password

METHOD	POST
ENDPOINT	`/api/auth/change-password/student`
DESCRIPTION	Change student password

```
{
  "id": 1,
  "currentPassword": "Test@123",
  "newPassword": "NewPass@123"
}
```

1.9 Change Teacher Password

METHOD	**POST**
ENDPOINT	`/api/auth/change-password/teacher`
DESCRIPTION	Change teacher password

Request Body: Same as change student password

2. OTP / Password Reset APIs

2.1 Send OTP (Forgot Password)

METHOD	**POST**
ENDPOINT	`/api/otp/forgot-password`
DESCRIPTION	Send OTP to email for password reset. Validates email exists for given role.

```
{
  "email": "student@college.edu",
  "role": "student"
}
```

role can be: student, teacher, admin

2.2 Verify OTP

METHOD	**POST**
ENDPOINT	`/api/otp/verify`
DESCRIPTION	Verify OTP code. OTP expires in 10 minutes and is single-use.

```
{
  "email": "student@college.edu",
  "otp": "123456"
}
```

2.3 Reset Student Password

METHOD	POST
ENDPOINT	`/api/auth/reset-password/student`
DESCRIPTION	Reset student password after OTP verification

```
{
  "email": "student@college.edu",
  "otp": "123456",
  "newPassword": "NewPass@123"
}
```

2.4 Reset Teacher Password

METHOD	POST
ENDPOINT	`/api/auth/reset-password/teacher`
DESCRIPTION	Same as reset student password

2.5 Reset Admin Password

METHOD	POST
ENDPOINT	`/api/auth/reset-password/admin`
DESCRIPTION	Same as reset student password

3. Student APIs

METHOD	ENDPOINT	DESCRIPTION
GET	`/api/students`	Get all students
GET	`/api/students/{id}`	Get student by ID

PUT	/api/students/{id}	Update student profile
DELETE	/api/students/{id}	Delete student (cascades attendance, marks, submissions)
PATCH	/api/students/{id}/status	Update student status (Active/Suspended/Graduated)
POST	/api/students/{id}/reset-password	Admin force reset student password

4. Teacher APIs

METHOD	ENDPOINT	DESCRIPTION
GET	/api/teachers	Get all teachers
GET	/api/teachers/{id}	Get teacher by ID
PUT	/api/teachers/{id}	Update teacher profile
DELETE	/api/teachers/{id}	Delete teacher (cascades courses, attendance, marks, etc.)
PATCH	/api/teachers/{id}/status	Update teacher status (Active/Suspended)

5. Course APIs

METHOD	ENDPOINT	DESCRIPTION
GET	/api/courses	Get all courses
GET	/api/courses/{id}	Get course by ID
GET	/api/courses/teacher/{teacherId}	Get courses by teacher
GET	/api/courses/department/{department}	Get courses by department
POST	/api/courses	Create course
PUT	/api/courses/{id}	Update course
DELETE	/api/courses/{id}	Delete course (cascades attendance, marks,

		assignments, materials)

6. Attendance APIs

METHOD	ENDPOINT	DESCRIPTION
GET	`/api/attendance/student/{studentId}`	Get attendance by student
GET	`/api/attendance/course/{courseId}`	Get attendance by course
GET	`/api/attendance/course/{courseId}/date/{date}`	Get attendance by course and date (YYYY-MM-DD)
GET	`/api/attendance/course/{courseId}/stats`	Get attendance stats (percentages per student)
POST	`/api/attendance`	Mark attendance (upsert)
PUT	`/api/attendance/{id}`	Update attendance record

7. Marks APIs

METHOD	ENDPOINT	DESCRIPTION
GET	`/api/marks/student/{studentId}`	Get marks by student
GET	`/api/marks/course/{courseId}`	Get marks by course
POST	`/api/marks`	Enter marks (auto-calculates grade)
PUT	`/api/marks/{id}`	Update marks
DELETE	`/api/marks/{id}`	Delete marks

8. Assignment APIs

METHOD	ENDPOINT	DESCRIPTION
GET	`/api/assignments`	Get all assignments
GET	`/api/assignments/course/{courseId}`	Get assignments by course
POST	`/api/assignments`	Create assignment
PUT	`/api/assignments/{id}`	Update assignment
DELETE	`/api/assignments/{id}`	Delete assignment

9. Course Materials APIs

9.1 Get Materials by Course

METHOD	GET
ENDPOINT	`/api/materials/course/{courseId}`

9.2 Upload Material (File)

METHOD	POST
ENDPOINT	`/api/materials/upload`
REQUEST	multipart/form-data

PARAMETER	TYPE	DESCRIPTION
courseId	Integer	Course ID
teacherId	Integer	Teacher ID
title	String	Material title
description	String	Optional description
file	File	File to upload

9.3 Add Material (Link)

METHOD	POST
ENDPOINT	/api/materials/link

```
{
  "courseId": "1",
  "teacherId": "1",
  "title": "Reference Notes",
  "description": "Optional",
  "url": "https://example.com/notes"
}
```

9.4 Download Material

METHOD	GET
ENDPOINT	/api/materials/download/{id}

9.5 Delete Material

METHOD	DELETE
ENDPOINT	/api/materials/{id}

10. Submission APIs

10.1 Submit Assignment (File Upload)

METHOD	POST
ENDPOINT	/api/submissions/upload
REQUEST	multipart/form-data

PARAMETER	TYPE	DESCRIPTION
assignmentId	Integer	Assignment ID
studentId	Integer	Student ID
file	File	Submission file

10.2 Get Submissions by Assignment

METHOD	GET
ENDPOINT	`/api/submissions/assignment/{assignmentId}`

10.3 Get Submissions by Student

METHOD	GET
ENDPOINT	`/api/submissions/student/{studentId}`

10.4 Check Submission Status

METHOD	GET
ENDPOINT	`/api/submissions/check?assignmentId={id}&studentId={id}`

10.5 Grade Submission

METHOD	PUT
ENDPOINT	`/api/submissions/{id}/grade`

```
{
  "grade": "A+",
  "comment": "Excellent work"
}
```

10.6 Download Submission

METHOD	GET
ENDPOINT	`/api/submissions/download/{submissionId}`

11. Health API

11.1 Database Health Check

METHOD	GET

ENDPOINT	/api/health/db

12. System Settings APIs

12.1 Get System Settings

METHOD	GET
ENDPOINT	/api/system/settings

12.2 Update System Settings

METHOD	PUT
ENDPOINT	/api/system/settings

13. Bulk Import APIs

13.1 Bulk Import Students

METHOD	POST
ENDPOINT	/api/bulk/students
REQUEST	multipart/form-data with CSV file

```
name,email,password,department,year,semester,phone,usn
Arjun Kumar,arjun@college.edu,Test@123,Computer Science and Engineering,3rd
```

13.2 Bulk Import Teachers

METHOD	POST
ENDPOINT	/api/bulk/teachers
REQUEST	multipart/form-data with CSV file

```
name,email,password,department,phone,employeeId
```

13.3 Bulk Import Attendance

METHOD	POST
ENDPOINT	`/api/bulk/attendance?courseId={id}&date={date}`
REQUEST	multipart/form-data with CSV file

CSV Format:

```
usn,status
1VT22CS001,Present
1VT22CS002,Absent
1VT22CS003,Present
```

13.4 Bulk Import Marks

METHOD	POST
ENDPOINT	`/api/bulk/marks?courseId={id}`
REQUEST	multipart/form-data with CSV file

```
usn,marks
1VT22CS001,85
1VT22CS002,72
1VT22CS003,91
```

API Summary

MODULE	TOTAL ENDPOINTS
Authentication	9
OTP / Password Reset	5
Students	6
Teachers	5

MODULE	TOTAL ENDPOINTS
Courses	7
Attendance	6
Marks	5
Assignments	5
Course Materials	5
Submissions	6
Health	1
System Settings	2
Bulk Import	4
Total	**65**